Environmental Economics and Finance

Markets, Measurement, and Policy

Sample Chapter: Chapter 1

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This is a sample chapter from a working draft. References to other chapters point to the full book. Figures labeled "stylized" are illustrations drawn for teaching and are not drawn to data.

Chapter 1

Introduction to Environmental Economics

Learning Objectives

After reading this chapter you should be able to:

1. Describe what the data show about pollution and economic growth in the United States and the world since 1970, and set those facts beside India's air and groundwater and Brazil's forests.
2. Explain why economists treat environmental problems as problems of allocating scarce resources, using scarcity, opportunity cost, marginal thinking and incentives.
3. Describe nature as an asset that yields a flow of services, and compute the present value of a simple flow.
4. Draw from a blank page the Micro 101 toolkit: demand as marginal benefit, supply as marginal cost, equilibrium, surplus and deadweight loss.
5. Build the marginal private cost and marginal social cost graph step by step, and show why the efficient level of pollution is usually not zero.
6. Separate positive from normative questions and efficiency from equity, and name the book's three threads: finance, water and climate.
7. Trace each U.S. lesson to India and Brazil: who owns the resource, who regulates it, where the data come from, and why the results differ.

Around noon on Sunday, 22 June 1969, an oil slick on the Cuyahoga River in Cleveland, Ohio, caught fire. Oily waste and debris had collected on the water, and the flames reached two railroad trestles over the river. Units from three fire battalions brought the fire under control in about twenty minutes (Encyclopedia of Cleveland History, 2026; Stradling and Stradling, 2008). The damage to the trestles came to $45,000 for the Norfolk & Western bridge and $5,000 for the Newburgh & South Shore trestle, in 1969 dollars (Encyclopedia of Cleveland History, 2026). The $50,000 total equals about $430,000 in 2024 dollars (U.S. Bureau of Labor Statistics, 2026).

The fire was not the river's first. The Cuyahoga had burned at least nine times before 1969 (Stradling and Stradling, 2008). A fire in 1952 did about $1.5 million of damage in 1952 dollars, about $18 million in 2024 dollars, and drew no national attention (Encyclopedia of Cleveland History, 2026).

The smaller fire of 1969 became a national symbol of polluted water and a rallying point that helped build support for the Clean Water Act, which Congress passed in 1972 (Phaneuf and Requate, 2016). Half a century later, Keiser and Shapiro (2019) take stock: public and private sources have spent nearly $5 trillion (2017 dollars) on clean rivers, lakes and drinking water since about 1970, water pollution has fallen, partly because of the laws, and yet over half of U.S. rivers still violate water-quality standards.

The Cuyahoga poses the questions of this book. The industries along the river used it as a free sewer, and the people who bore the cost were not the people who caused it. How clean should a river be? Who should pay? Which rules work, and at what cost? Environmental economics answers these questions with the tools of an introductory economics course. This chapter sets out the facts, builds the tools one graph at a time, and reports what the evidence says.

India and Brazil

Clean Water Act. India: Water (Prevention and Control of Pollution) Act, 1974. Brazil: National Water Resources Policy, Law 9.433 of 1997.

1.1 What the Data Show

E340 Lecture 1

The facts start in the United States, where the data are long and the record of policy is well studied. Each U.S. fact then has a counterpart in India and in Brazil, and the counterparts differ in revealing ways.

1.1.1 Fact 1: U.S. air pollution fell while the economy grew

Since 1970 the U.S. economy has grown more than fourfold while the main air pollutants have fallen. Between 1970 and 2025, real gross domestic product (GDP) rose 4.5-fold (U.S. Bureau of Economic Analysis, 2026). Over the same years, national emissions of sulfur dioxide (SO_2) fell 95 percent, carbon monoxide 78 percent and nitrogen oxides 71 percent (U.S. Environmental Protection Agency, 2026b). Figure 1.1 shows each series as an index equal to 100 in 1970.

Definition: Decoupling

Growth in output without a matching growth in pollution. Pollution can fall in absolute terms (absolute decoupling) or only per dollar of output (relative decoupling).

The fall in SO_2 is the largest. Emissions went from about 23 million short tons in 1990 to 1.6 million in 2025 (U.S. Environmental Protection Agency, 2026b). The steepest part of the decline came after the 1990 Clean Air Act Amendments, which capped power-plant SO_2 and let plants trade the right to emit (Chapter 6, Chapter 20). Economists call the pattern *decoupling*: output grows while pollution falls.

Carbon dioxide does not follow the other pollutants. U.S. fossil CO_2

India and Brazil

Clean Air Act. India: Air (Prevention and Control of Pollution) Act, 1981. Brazil: national air-quality standards set by CONAMA, Resolution 491 of 2018.

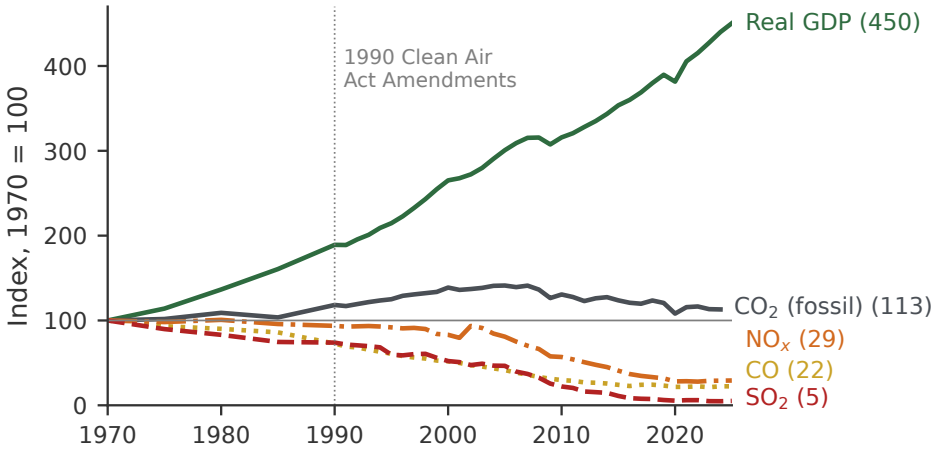


Figure 1.1 — U.S. real GDP and national emissions of four pollutants, 1970–2025, each indexed to 1970 = 100; the number after each label is the latest index value. Sulfur dioxide (SO_2), nitrogen oxides (NO_x) and carbon monoxide (CO) exclude wildfires and are reported at five-year intervals before 1990. Carbon dioxide (CO_2) is fossil-fuel and industrial emissions through 2024. Sources: U.S. Environmental Protection Agency (2026b); U.S. Bureau of Economic Analysis (2026); Friedlingstein et al. (2026) via Our World in Data (2026).

emissions in 2024 were 13 percent above their 1970 level, though 20 percent below their 2005 peak (Friedlingstein et al., 2026). The gap between the falling SO_2 line and the flat CO_2 line in Figure 1.1 is one of the puzzles this chapter explains.

1.1.2 Fact 2: world carbon dioxide emissions are still rising

World fossil CO_2 emissions rose from 14.9 billion tonnes (Gt) in 1970 to 38.6 Gt in 2024, a 2.6-fold increase (Friedlingstein et al., 2026). (The Global Carbon Budget also reports a slightly lower total, 37.8 Gt for 2024, which subtracts the CO_2 that cement reabsorbs as it ages; the figures here are before that subtraction.) Per person, the world average is 4.7 tonnes a year, about what a typical U.S. passenger car emits in a year, 4.6 tonnes (U.S. Environmental Protection Agency, 2026c). Figure 1.2 splits the total by emitter.

The growth since 2000 has come from outside the rich countries. Emissions from the United States and the European Union have fallen from their peaks. China emitted 12.3 Gt in 2024, 32 percent of the world total; the United States emitted 4.9 Gt, or 13 percent (Friedlingstein et al., 2026).

Totals hide large differences per person, and Figure 1.3 maps them.

Definition: Greenhouse gas

A gas, such as CO_2 or methane, that traps heat in the atmosphere. It mixes around the globe, so a ton emitted in Indiana warms the earth as much as a ton emitted in India.

Example

An average U.S. resident accounts for 14.2 tonnes of fossil CO_2 a year, about three times the world average of 4.7 tonnes.

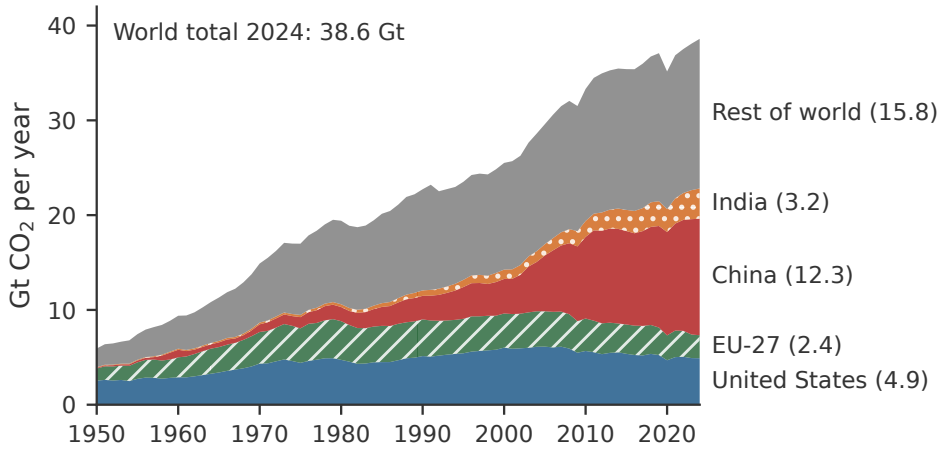


Figure 1.2 — Fossil-fuel and industrial CO_2 emissions by emitter, 1950–2024, in billion tonnes of CO_2 per year, stacked; excludes land-use change. The number after each label is the 2024 value. EU-27: the 27 current members of the European Union. Source: Global Carbon Budget 2025 (Friedlingstein et al., 2026), via Our World in Data (2026).

In 2024 an average resident of the United States emitted 14.2 tonnes, of China 8.7 tonnes, of Brazil 2.3 tonnes, of India 2.2 tonnes and of Ethiopia 0.14 tonnes; at the top, Qatar emitted about 41 tonnes per person (Friedlingstein et al., 2026). About 64 percent of the world's people live in countries below the world average of 4.7 tonnes. The map previews a question for Chapter 15 and Chapter 27: the countries whose emissions are growing fastest are still far below the rich countries per person, so who should cut, and who should pay?

Per-person emissions also depend on what is counted (Figure 1.4). In 2024 an average resident of India emitted 2.2 tonnes of fossil CO_2, 3.3 times the 0.7 tonnes of 1990, and an average Brazilian about as much, 2.3 tonnes (Friedlingstein et al., 2026). Counting the CO_2 released by clearing forests raises Brazil's figure to 9.8 tonnes: land-use change made up about 77 percent of Brazil's CO_2 emissions, 1.6 Gt against 483 million tonnes from fossil fuels (Friedlingstein et al., 2026). India's and the United States' totals barely move when land is counted. Brazil's per-person total peaked at about 19 tonnes in 2003 and fell as Amazon clearing slowed after 2004, a decline Section 1.1.4 returns to.

Example

An average Ethiopian would need about 110 years to emit what an average U.S. resident emits in one year (14.2 versus 0.14 tonnes).

Key Point

A carbon price on fossil fuels reaches most U.S. and Indian CO_2. It misses most of Brazil's, which comes from clearing land.

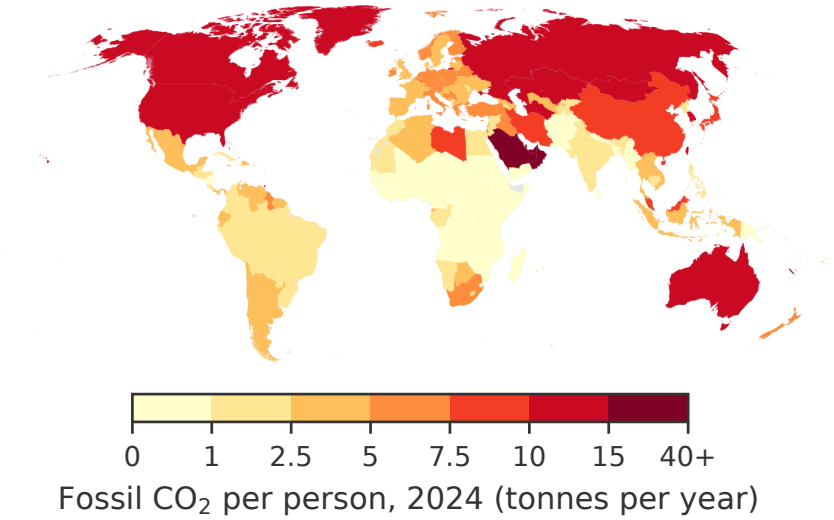


Figure 1.3 — Fossil-fuel and industrial CO_2 emissions per person by country, 2024, in tonnes of CO_2 per person per year; excludes land-use change. Light grey: no data. Robinson projection; country outlines from Natural Earth. Source: Global Carbon Budget 2025 (Friedlingstein et al., 2026), via Our World in Data (2026).

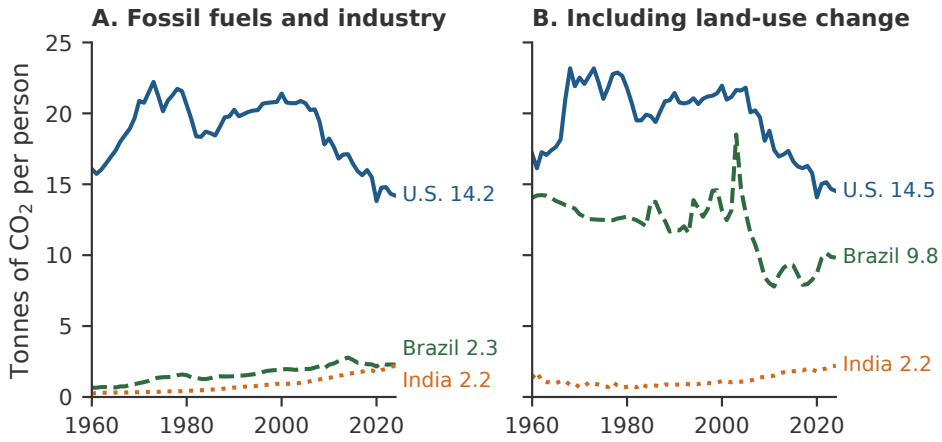


Figure 1.4 — Carbon dioxide emissions per person in the United States, India and Brazil, 1960–2024, in tonnes of CO_2 per person per year; the number after each label is the 2024 value. Panel A: fossil fuels and industry (as in Figure 1.3). Panel B: fossil fuels and industry plus land-use change, mainly the clearing of forests. Land-use-change emissions come from bookkeeping models and are far less certain than fossil emissions; the year-to-year swings in Brazil's line are partly model noise. Source: Global Carbon Budget 2025 (Friedlingstein et al., 2026), via Our World in Data (2026).

1.1.3 Fact 3: water problems have not gone away

Rivers no longer burn, but water pollution still reaches drinking water. In August 2014, a toxic algae bloom in western Lake Erie, fed by phosphorus that ran off farm fields, left Toledo, Ohio, without drinking water for more than two days (Steffen et al., 2017). In Flint, Michigan, a 2014 switch of the city's water source let lead leach from old pipes. The share of young children with elevated blood lead rose from 2.4 to 4.9 percent, roughly doubling, with the largest increases in the poorest neighborhoods (Hanna-Attisha et al., 2016).

Definition: Nonpoint source

Pollution from many diffuse places, such as runoff from thousands of farm fields, rather than from one pipe (a *point source*). It is hard to measure and hard to regulate.

1.1.4 Fact 4: India's air and groundwater, Brazil's forests

Air pollution is India's largest environmental health problem. The Global Burden of Disease study attributes 1.67 million deaths in India in 2019 to air pollution, 17.8 percent of all deaths in the country, and estimates the lost output from those deaths and illnesses at 1.36 percent of India's GDP (India State-Level Disease Burden Initiative, 2021). For comparison, the 1990 Clean Air Act Amendments were projected to avoid about 230,000 early U.S. deaths a year by 2020 (U.S. Environmental Protection Agency, 2011).

Example

1.67 million deaths a year is about 4,600 a day. The same study finds that death rates from household smoke fell by about two-thirds from 1990 to 2019, while those from outdoor particles more than doubled (India State-Level Disease Burden Initiative, 2021).

India's groundwater problem is mainly one of quantity: the water is being used up. Satellites that weigh the water underground find that northwest India, the states of Rajasthan, Punjab and Haryana with Delhi, lost about 109 cubic kilometers of groundwater from August 2002 to October 2008, while rainfall stayed near normal; pumping for irrigation is the likely cause (Rodell et al., 2009). Worked Example 1.2 turns the number into a stock and a flow.

Brazil's largest environmental problem is the clearing of forest. INPE, Brazil's national space research institute, measured 6,518 km^2 of clear-cut forest in the Legal Amazon in 2024 (Instituto Nacional de Pesquisas Espaciais, 2025), about 18 km^2 a day (= 6,518/365). The rate of Amazon deforestation fell by 84 percent between 2004 and 2012 (Heilmayr et al., 2020), and Figure 1.4 shows the fall in Brazil's emissions per person. Chapter 25 tells that story; the economics of why it happened begins in Section 1.6.

Definition: Land-use change emissions

CO_2 released when forests and other vegetation are cleared or burned, net of regrowth. They are estimated with models and are much less certain than fossil emissions.

Trace It: India and Brazil — Who measures the facts

Institution

U.S.	EPA (1970) compiles the emissions record of Figure 1.1 (U.S. Environmental Protection Agency, 2026d).
India	The CPCB, constituted in 1974, with state pollution control boards (Central Pollution Control Board, 2026).
Brazil	INPE maps forest clearing; CONAMA, created by Law 6.938 of 1981, sets pollution standards (Brasil, 1981).

Rule or market

U.S.	The Clean Air Act's national air-quality standards (Chapter 20).
India	The Air Act of 1981 lets the CPCB set air-quality standards (Central Pollution Control Board, 2026).
Brazil	The national air-quality policy law of 2024 (Law 14.850) and CONAMA Resolution 506 of 2024 set air standards (Brasil, 2024; Conselho Nacional do Meio Ambiente, 2024); satellite alerts target forest enforcement (Assunção et al., 2023).

Data

U.S.	EPA emissions trends; BEA output.
India	Ground monitors and modeled exposure (India State-Level Disease Burden Initiative, 2021); satellite gravity for groundwater (Rodell et al., 2009).
Brazil	PRODES annual maps and DETER alerts (Instituto Nacional de Pesquisas Espaciais, 2025).

Evidence and difference

U.S.	Long, dense records make U.S. trends easy to read.
India	Monitoring is costly and can be bent: in Gujarat, auditors paid by the plants they inspected reported emissions just below the standard when true emissions were typically higher (Duflo et al., 2013).
Brazil	Clearing is visible from space at low cost, so Brazil sees its largest problem better than India sees its air.

Why did SO_2 fall and CO_2 rise? Why did the Cuyahoga burn for decades before anyone acted? The answers start with how economists think about scarce resources.

1.2 The Economic Way of Thinking

E340 Lecture 1

1.2.1 Economics: choices under scarcity

Economics studies how people, firms and governments choose when they cannot have everything they want. A family chooses between a newer car and a vacation. A city chooses between a new water plant and more police officers. A country chooses how much coal to burn this year. Four ideas from your principles course carry most of this book.

The first idea is *scarcity*. Land, labor, time, clean air and fresh water are limited, and the uses people would like to put them to are not. Clean air is scarce in this sense. The air can absorb only so much smoke before people get sick, and every factory and car would like to use it as a free dump.

The second idea is *opportunity cost*: the value of the best thing you give up when you choose. Keeping a forest standing gives up the timber or houses the land could have produced. Cutting it gives up the carbon storage, wildlife habitat and flood control the forest provided. Opportunity cost is not the same as cash spent. A town that leaves a wetland undeveloped spends nothing, but it gives up what a developer would have paid for the land.

The third idea is *marginal thinking*. Most choices are about a little more or a little less, not all or nothing. A power plant does not choose between polluting and not polluting. It chooses how many tons of SO_2 to remove from its smoke, and each extra ton costs more to remove than the last. The rule follows directly: keep doing more of something as long as one more unit adds more benefit than cost, and stop when the two are equal. That one rule produces most of the results in this book.

The fourth idea is *incentives*. People respond to prices, rules and information. When gasoline costs more, people drive less and buy more efficient cars. When a firm pays a fee for every ton it emits, it looks for cheap ways to emit less. Much of environmental policy is the design of incentives that line up private choices with what is good for everyone.

Definition: Economics

The study of how individuals, firms and governments choose among alternatives when resources are scarce, and what those choices mean for society.

Definition: Scarcity

Having fewer resources than the uses people want to put them to. Scarcity forces choice.

Definition: Opportunity cost

The value of the best alternative given up by a choice. It is the true cost of an action, even when no money changes hands.

Definition: Marginal

"One more unit." The marginal benefit (cost) of an action is the extra benefit (cost) of doing one more unit of it.

Definition: Incentive

Anything that changes the costs or benefits a person faces (a price, tax, fine, subsidy, rule or piece of information) and so changes behavior.

Micro 101 Refresher: Scarcity and the production possibilities frontier

A *production possibilities frontier* (PPF) shows the combinations of two goods an economy can produce when it uses all its resources well. Points on the frontier are *productively* efficient (neither good can be increased without giving up some of the other), points inside it waste resources, and points outside it are out of reach. The slope of the frontier is the opportunity cost of one good in terms of the other. The frontier usually

bows outward, so the opportunity cost of a good rises as the economy produces more of it.

The PPF is the simplest honest picture of the environmental trade-off. Put "market goods" (everything bought and sold) on one axis and "environmental quality" (clean air, clear lakes, standing forests) on the other.

Definition: Production possibilities frontier (PPF)

The boundary of what an economy can produce with its resources and technology. Its slope is the opportunity cost of one good in terms of the other.

Build the Graph Step by Step: The environment–economy trade-off (PPF)

Step 1. Draw the axes. Put market goods G on the horizontal axis and environmental quality Q on the vertical axis. Both are goods: more is better.

Step 2. Draw the frontier bowed outward, from the vertical intercept (all resources to environmental quality) to the horizontal intercept (all resources to market goods).

Step 3. Mark two points on the frontier: A (more goods, less environmental quality) and B (fewer goods, more environmental quality). The goods given up between A and B are the opportunity cost of the cleaner environment.

Step 4. Mark a point C inside the frontier. At C the economy could have more of both goods; costly, badly designed pollution control puts an economy there.

Step 5. Mark a point D outside the frontier, out of reach today. Draw a second, larger frontier to show how better technology (cheaper scrubbers, cheaper solar panels) brings D within reach.

Worked Example 1.1: Reading opportunity cost off a PPF

An economy's frontier passes through $(G, Q) = (10, 0)$, $(8, 6)$, $(6, 8)$ and $(0, 10)$, where G is billions of dollars of market goods and Q is an index of environmental quality.

1. From $(10, 0)$ to $(8, 6)$: give up 2 of G, gain 6 of Q. Each unit of Q costs $2/6 \approx 0.33$ of goods.
2. From $(8, 6)$ to $(6, 8)$: give up 2, gain 2. Each unit of Q costs $2/2 = 1$.
3. From $(6, 8)$ to $(0, 10)$: give up 6, gain 2. Each unit of Q costs $6/2 = 3$.

The opportunity cost of environmental quality rises ninefold, from 0.33 to 3, as the economy asks for more of it. The first cleanups are cheap and the last are expensive, the same logic as the rising cost of abatement

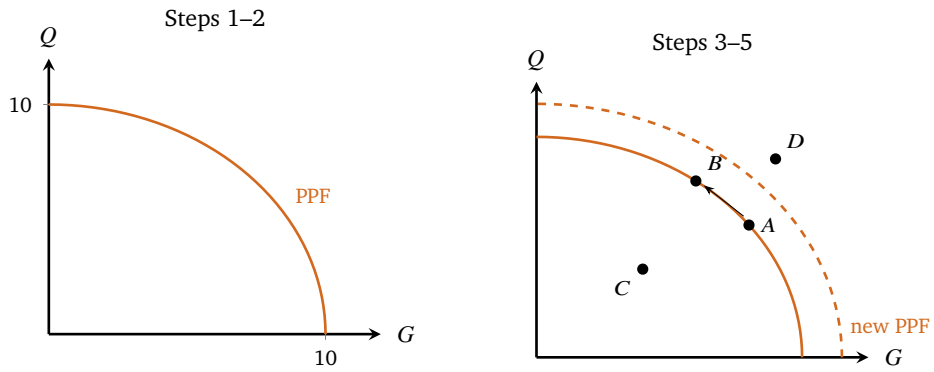


Figure 1.5 — The production possibilities frontier for market goods G and environmental quality Q, built in steps. Left: the bowed-out frontier. Right: the move from A to B trades goods for environmental quality; C wastes resources; D is out of reach until better technology shifts the frontier out (dashed). *(Stylized illustration; not drawn to data.)*

in Section 1.7.

The frontier carries two lessons. On the frontier, environmental protection costs other things people value; a claim that a rule has "no cost" assumes the economy sits inside the frontier. Being on the frontier does not say which point is best: A and B are both productively efficient, and which one society prefers depends on how people value market goods relative to environmental quality. Many real economies do sit inside it, because they control pollution in needlessly expensive ways. Part II of this book is largely about moving from points like C toward the frontier.

Key Point

On the frontier, more environmental quality costs market goods. Inside it, better policy can deliver more of both.

1.2.2 Environmental problems are allocation problems

Environmental economics applies these four ideas to the environment. Smog, algae blooms and a warming climate are physical events, but they follow from human choices about what to produce, what to buy and where to put the waste. How much they matter depends on what people value.

Definition: Environmental economics

The use of economic reasoning to study why environmental problems arise, how to value environmental goods, and how to design policies that improve outcomes.

A chemist explains a burning river with oil and sparks. An economist asks why the industries along the Cuyahoga let oily waste reach it. The answer is that dumping was free to them, while the damage fell on others: boat owners, anglers, the railroads whose trestles burned, and everyone downstream. A scarce resource, the river's capacity to absorb waste, was handed out without a price, so it was overused.

Example

An Ohio coal plant's SO_2 falls as acid rain on lakes in New York. The plant's customers get the power; anglers downwind bear the cost. No one in the sale speaks for them.

Phaneuf and Requate (2016) put the starting point of the field in the same terms: one person's use of the environment directly affects someone else, without that person's consent and without payment. A factory's smoke soils the laundry next door; nitrogen fertilizer from a farm seeps into a town's wells. Markets handle such cases badly, and they are the subject of this book. Four questions organize it, and they are the four questions of E340:

1. **Why do environmental problems arise?** When do private choices produce bad outcomes for society? (Part I.)
2. **How are costs and benefits measured?** How can clean air, clean water and healthy ecosystems be valued when no one buys or sells them? (Part III.)
3. **Which policies work?** Regulations, taxes, tradable permits, subsidies, property rights and information. (Parts II and V.)
4. **Who gains, who pays, and is the outcome fair?** (Part VI, and throughout.)

1.2.3 Positive and normative questions

Economists separate two kinds of statements. A *positive* statement says how the world is. "A tax of \$31 per ton of CO_2 (2010 dollars) would raise the price of gasoline by about 28 cents a gallon" is positive: a gallon of gasoline releases 8,887 grams of CO_2 (U.S. Environmental Protection Agency, 2026c), and $0.008887 \times 31 \approx 0.28$. Evidence can check it.

Definition: Positive analysis

Statements about what is, was or will be: cause and effect, predictions and measurements. Evidence can settle them.

A *normative* statement says how the world ought to be. "The United States should adopt a carbon tax" is normative. It depends on values: how much weight future generations get, how much weight people in other countries get, and who should pay higher energy bills.

Definition: Normative analysis

Statements about what ought to be. They combine facts with value judgments about which outcomes are better.

Most public arguments about the environment mix the two. When two people disagree about a rule, ask whether they disagree about facts (how many asthma attacks the rule prevents) or about values (what an avoided asthma attack is worth, and who should pay). Economics has much to say about facts; about values, it can make the trade-offs explicit but cannot settle them.

1.2.4 Three threads: finance, water and climate

Three threads run through this book, and each appears in this chapter.

Key Point
The book's three threads: the environment as a financial asset, water, and climate.

Environment and finance. Forests, aquifers, fisheries and a stable climate are assets that pay a stream of services, as a bond pays interest. The tools of finance (present value, depreciation, risk, discounting) value those assets and show when they are being run down. Section 1.3 starts this thread, and Part IV develops it with climate finance, green bonds and natural capital accounting. Brazil's Amazon Fund is a finance case of its own: it pays Brazil for reductions in Amazon deforestation and, in 2009, valued each tonne of CO_2 kept in the forest at $5 (BNDES, 2010); Chapter 17 explains how.

Water. The second thread is the quality, safety and scarcity of water: the Cuyahoga, Lake Erie and Flint, and also the price of water. Many cities charge far less for water than it costs to supply in a drought, so households have little reason to conserve, and pricing can save water at lower cost than rationing (Keohane and Olmstead, 2016). In much of India the price problem is starker: farmers pay a flat fee or nothing for the electricity that runs their pumps, so the last liter of groundwater costs them almost nothing to lift (Ryan and Sudarshan, 2022). Chapter 21 (Clean Water Act), Chapter 22 (Safe Drinking Water Act) and Chapter 18 (India's groundwater) develop this thread.

Climate. The third thread is climate change. Heat lowers learning: students learn less in hotter school years (Park et al., 2020). Extreme heat also raises death rates, most in poorer and hotter places (Carleton et al., 2022), and in Indian factories hot days lower workers' output (Somanathan et al., 2021). In the Amazon, climate and land use interact: Gong et al. (2026) ask whether droughts push landholders to clear more forest. Economists summarize the damage from one more ton of CO_2 in a single number, the *social cost of carbon*. Nordhaus (2017) estimates it at $31 per ton of CO_2 for 2015, in 2010 dollars.

Definition: Social cost of carbon (SCC)
The dollar value of all future damage caused by emitting one more ton of CO_2 today, discounted to the present.

For a typical car emitting 4.6 tonnes a year (U.S. Environmental Protection Agency, 2026c), that is about $140 a year of damage to others ($= 4.6 \times 31$).

1.3 The Environment as an Asset

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The "finance" in this book's title starts from one idea: nature is a form of wealth. A forest, an aquifer, a fishery and a clean river are assets.

Like a factory or a government bond, each is a stock that pays out a stream of services over time: clean water, stored carbon, pollinated crops, buffered storms, fish, timber and recreation. These services have value even when no one pays a price for them.

Definition: Natural capital

The stock of natural assets (forests, soils, fisheries, aquifers, minerals, the atmosphere, biodiversity) that provides a flow of goods and services to people.

1.3.1 Stocks and flows

A stock is an amount at a point in time; a flow is an amount per unit of time. Your savings balance is a stock and the yearly interest is a flow; the fish in a lake are a stock and the yearly catch is a flow.

Picture a bathtub: the tap (growth, restoration) adds to the stock, the drain (harvest, depletion, degradation) takes away, and the level changes only when the two differ. Overfishing happens when the catch exceeds the fish population's natural growth year after year. Climate change is a stock problem of the same kind: even if yearly CO_2 emissions fell, the amount in the atmosphere would keep rising as long as emissions exceed what oceans and forests absorb.

Definition: Stock vs. flow

A stock is measured at a moment (tons of fish in a lake on 1 January). A flow is measured per period (tons caught per year). Flows add to or subtract from stocks.

Definition: Ecosystem services

The benefits people get from nature: provisioning (food, timber, water), regulating (flood control, climate, pollination), cultural (recreation) and supporting (nutrient cycling).

Worked Example 1.2: Groundwater in northwest India as a stock and a flow

Rodell et al. (2009) use two gravity-measuring satellites to weigh the water under Rajasthan, Punjab and Haryana (with Delhi). From August 2002 to October 2008, about 6.2 years, the region lost about 109 cubic kilometers (km^3) of groundwater, roughly twice the capacity of India's largest surface reservoir.

1. *The flow.* $109/6.2 \approx 17.6$ km^3 a year, close to the authors' estimate of 17.7 ± 4.5 km^3 a year.
2. *The bathtub.* The stock falls by the drain minus the tap: pumping minus recharge. A net loss of about 17.7 km^3 a year means pumping exceeded recharge by that much, every year. The level stops falling only when pumping falls or recharge rises by that amount.
3. *Depth.* Spread over the whole region, the loss equals a layer of water

about 4 centimeters deep each year (Rodell et al., 2009). Wells must go deeper, and each meter of extra depth raises the energy needed to lift each liter.

4. *Why the drain stays open.* Where farm electricity is free or charged at a flat rate, the farmer's marginal cost of pumping is close to zero (Ryan and Sudarshan, 2022). Each farmer weighs the value of water against his own pumping cost and ignores the fall in his neighbors' water table, the common-pool problem of Section 1.6.

A falling stock with an unchanged flow is the signature of an asset being run down. Counted at its value, the lost water is depreciation of India's natural capital, the subject of Chapter 18.

1.3.2 What an asset is worth

An asset is worth the present value of the benefits it will provide. A dollar next year is worth less than a dollar today, because today's dollar can earn interest; future benefits are therefore *discounted*. For an asset that pays the same amount B every year forever, the formula is short:

$$PV = \frac{B}{r}, \tag{1.1}$$

where r is the yearly discount rate. The formula assumes the benefit is the same every year forever and the rate never changes; if the benefit grows, shrinks or ends, the value changes. Chapter 10 treats discounting in full.

Definition: Present value (PV)
What a stream of future amounts is worth today. Each future amount is discounted at rate r: $PV = \sum_t B_t/(1+r)^t$.

Worked Example 1.3: What is a wetland worth?

A U.S. county owns a wetland upstream of a small town. The wetland soaks up floodwater and cuts the town's expected flood damage by \$60,000 a year (illustrative numbers). A developer offers \$1.5 million for the land.

1. *Value as an asset.* At $r = 3\%$, $PV = 60{,}000/0.03 = \$2{,}000{,}000$.
2. *Decision.* The sale brings in \$1.5 million and destroys flood protection worth \$2.0 million, so society is \$0.5 million poorer before counting recreation or wildlife.
3. *Damage as depreciation.* If silt cuts the wetland's protection to \$45,000 a year, its value falls to $45{,}000/0.03 = \$1{,}500{,}000$. The

$500,000 loss is depreciation of natural capital, although nothing was sold.

4. *The discount rate matters.* At $r = 6\%$, the intact wetland is worth $1,000,000, and the developer's offer looks good.

Worked Example 1.4: What a year of Amazon clearing costs

The same accounting applies to a forest, whose lost service is carbon storage. INPE measured 6,518 km^2 of clear-cut forest in the Legal Amazon in 2024 (Instituto Nacional de Pesquisas Espaciais, 2025). Brazil's Amazon Fund assumes 100 tonnes of carbon per hectare, or 367 tonnes of CO_2, and valued each tonne of CO_2 at $5 (BNDES, 2010).

1. *Area.* One km^2 is 100 hectares, so $6{,}518 \times 100 = 651{,}800$ hectares.
2. *Carbon released.* $651{,}800 \times 367 \approx 239$ million tonnes of CO_2, about half of Brazil's 483 million tonnes of fossil CO_2 in 2024 (Friedlingstein et al., 2026).
3. *Value at the fund's price.* 239 million $\times$ \$5 $\approx$ \$1.2 billion.
4. *Value at the social cost of carbon.* At \$31 per tonne (2010 dollars; Section 1.2), 239 million $\times$ 31 $\approx$ \$7.4 billion.

Like the silted wetland, the lost forest is depreciation that GDP does not record, while the pasture that replaces it counts as output. The two prices differ sixfold, so whether clearing "pays" depends on the price used and on who pays it.

The asset view exposes a blind spot in standard statistics. GDP counts the developer's construction spending as output, and nothing subtracts the lost flood protection. A country can raise measured income for years by running down its natural capital, as a family can live well for a while by spending its savings. The U.K. Treasury's review of the economics of biodiversity estimates that, between 1992 and 2014, produced capital per person doubled worldwide while natural capital per person fell by nearly 40 percent (Dasgupta, 2021).

Key Point

An environmental asset is worth the present value of its services. Degrading it is depreciation, even when no market records the loss.

Finance Connection: Nature as a portfolio

An analyst asks three questions about a company: What assets does it own? What do they earn? Is it maintaining them, or raising this

year's profit by running them down? The same questions apply to a country's natural capital. A fishery is a bond whose interest is the sustainable catch; overfishing spends the principal. National accounts subtract the wear on machines but not depleted aquifers (Chapter 18). Environmental risk also shows up in asset prices (Part IV).

From the Research: What water pollution costs the United States: Keiser et al. (2025)

If nature is an asset, pollution is a hidden charge against it. David Keiser, Daniel Phaneuf, Catherine Kling and coauthors, among them the author of this book, put a price on it for nutrient pollution (nitrogen and phosphorus) from farms, cities and wastewater plants (Keiser et al., 2025). Their national model traces pollution through about 2.5 million river segments and more than 350,000 lakes, and converts changes in water quality into dollars of lost recreation, home value, drinking-water treatment, health and climate damage. With 2012 emissions, the baseline estimate is about $53 billion a year (2022 dollars), roughly 0.25 percent of U.S. GDP, within a range of about $35–100 billion. Lost recreation accounts for about 88 percent of the damage. Farms supply about two-thirds of the nutrients, yet municipal wastewater plants, which discharge close to where people live, contribute more of the damage. Chapter 21 returns to the study.

1.3.3 Sustainability: keeping the capital stock intact

The asset view leads to sustainability. The Brundtland Commission defines sustainable development as development that meets present needs without compromising the ability of future generations to meet theirs (World Commission on Environment and Development, 1987). Few people disagree with the definition; the difficulty lies in measuring it.

Definition: Sustainable development

Development that meets present needs without compromising the ability of future generations to meet theirs (Brundtland Commission, 1987).

Solow (1993) gives an economic reading. Sustainability means keeping the economy's capacity to produce well-being, not keeping every particular resource. Using up oil is consistent with sustainability if each generation replaces what it uses with other assets of at least equal value: machines, knowledge, skills, restored ecosystems. In the language of finance, a society should live off the interest and not the principal. Solow ties the idea to Hartwick (1977), who shows that a society that invests all its resource rents

Definition: Hartwick rule

Invest the rents (net revenues) from using up exhaustible resources in other capital; consumption can then be held constant over time.

in other capital can keep consumption from falling. On this reading the mistake is not mining but spending the proceeds of mining. The result comes from a simple model: it assumes that produced capital can stand in for the resource used up and that resource rents are measured at the right prices. Strong sustainability, below, rejects the first assumption for some assets.

Solow also asks for better measurement. Net national product (NNP) subtracts the wear on machines from GDP but ignores the running down of nature. Subtracting resource depletion and environmental damage as well gives a "green" NNP, which shows whether a country is living off its interest. Like the Hartwick rule, it is a guide rather than an exact test: it is only as reliable as the prices used to value depletion and damage.

Formula: green NNP

Green NNP = GDP – depreciation of produced capital – resource depletion – environmental degradation.

Worked Example 1.5: Is this economy sustainable?

A country reports, in billions of dollars a year: GDP = 1,000; gross investment in produced capital = 150; depreciation of produced capital = 100; resource rents from minerals extracted = 60; environmental degradation = 40 (illustrative numbers).

1. Net national product: 1,000−100 = 900. Green NNP: 900−60−40 = 800.
2. Net investment in produced capital: 150 – 100 = 50.
3. Change in total capital: 50 – 60 – 40 = −50, or 5 percent of GDP.

Total capital shrinks by $50 billion a year, so the economy fails the Solow–Hartwick test even if GDP grows. It would need $50 billion more net investment a year, or less depletion, to pass.

Solow's view is called *weak sustainability* because it lets produced capital substitute for natural capital.

Definition: Weak sustainability

The total value of all capital (produced, human, natural) must not decline. Natural capital may be run down if other capital replaces it.

Strong sustainability rejects that substitution for some assets. A stable climate, a species or the ozone layer may have no adequate substitute and must be preserved in physical terms (Tol, 2024). In the worked example, a strong-sustainability advocate objects to the $100 billion loss of natural capital even if produced capital grows enough to offset it.

Definition: Strong sustainability

Certain stocks of natural capital must be maintained in physical terms, because other capital cannot replace them.

From the Research: Rebuilding natural capital: Baragwanath et al. (2023)

Natural capital can grow as well as shrink. Baragwanath et al. (2023) ask whether secure land rights help forests regrow in Indigenous territories

of the Brazilian Amazon. Earlier work by Baragwanath and Bayi (2020) finds that full legal rights ("homologation") reduce deforestation inside the territories. Using satellite measures of secondary forest, a regression discontinuity design at territory boundaries and a staggered difference-in-differences design based on the timing of homologation, they find that full collective property rights increase both the extent and the age of secondary forest: at territory borders, land inside has about 5 percentage points more secondary forest after homologation, and its secondary forests are about 2.2 years older. Chapter 3 and Chapter 25 return to the study.

1.4 The Micro 101 Toolkit I: Demand, Supply and Equilibrium

E340 Lecture 1 (micro recap)

Introductory microeconomics explains how markets work when nothing goes wrong. Environmental economics studies what happens when something does, so the first step is a clear picture of a market that works. The next two sections rebuild that picture one line at a time, as in an E340 lecture. Every graph later in the book is a variation on these lines.

One market runs through the chapter: electricity in a regional wholesale market. Quantity is thousands of megawatt-hours (MWh) a day; price is dollars per MWh, and all prices in the example are illustrative. Generating electricity is one of the largest sources of air pollution and greenhouse gases in the United States.

Definition: Market

All the actual and potential buyers and sellers of a good, whose trades set its price and quantity.

1.4.1 Demand is marginal benefit

Micro 101 Refresher: Demand = marginal benefit = willingness to pay

A *demand curve* shows the quantity buyers want at each price. Read it two ways. *Horizontally*: at price P, how much do buyers want? *Vertically*: at quantity Q, what is the most the last buyer would pay for one more unit? That amount is the buyer's *marginal willingness to pay*, which economists treat as the *marginal benefit* of the good. Demand slopes down because each extra unit is worth less than the one before, and because a higher price sends buyers to substitutes and leaves them less

to spend.

The vertical reading matters most in environmental economics. If the 20,000th MWh is worth \$80 to whoever buys it, society gains \$80 of benefit when that MWh is produced and used. Adding willingness to pay over all units gives the total benefit. When a good has no market, such as clean air or a clear lake, Part III shows how to recover the same demand curve by other means.

Definition: Willingness to pay (WTP)

The most a person would give up to get a good or an improvement. It measures the good's benefit to that person in dollars.

Market demand adds up the quantities each buyer wants at each price. In the electricity market, suppose demand is

$$P = 120 - 2Q,$$

so the first unit is worth up to \$120 per MWh to the buyer who values it most, and willingness to pay falls by \$2 for each additional thousand MWh.

Board tip

Sum private-good demands horizontally: fix a price and add the quantities. For public goods, Chapter 2 sums vertically instead.

1.4.2 Supply is marginal cost

Micro 101 Refresher: Supply = marginal cost

A *supply curve* shows the quantity sellers want to sell at each price. Read vertically, it shows the *marginal cost* of the last unit, the lowest price a seller would accept. A competitive firm produces until price equals its marginal cost, so market supply is the horizontal sum of firms' marginal cost curves. Supply slopes up because producers use their cheapest resources first.

The upward slope is visible in how a power grid runs. Grid operators call on the cheapest plants first (wind, solar, nuclear), then efficient gas plants, and last the old "peaker" plants that run only on the hottest days. The cost of the last MWh therefore rises with output. Suppose market supply is

$$P = 20 + 2Q.$$

Definition: Marginal cost (MC)

The extra cost of producing one more unit. A competitive firm's supply curve is its marginal cost curve.

The supply curve leaves out any cost the plants impose on others without paying for it. Section 1.6 returns to that omission, and for this reason supply is often labeled *MPC*, the *marginal private cost*.

Build the Graph Step by Step: Demand, supply and equilibrium

Step 1. Draw the axes: quantity Q (thousand MWh a day) on the horizontal axis, price P ($ per MWh) on the vertical axis. Label both; on an exam, unlabeled axes lose points.

Step 2. Draw demand as a straight line from $120 on the vertical axis to $Q = 60$ on the horizontal axis. Label it $D = MB$. Practice both readings at one point: at $80 buyers want 20 (horizontal); the 20th unit is worth at most $80 to its buyer (vertical).

Step 3. Draw supply as an upward-sloping line starting at $20 on the vertical axis. Label it $S = MPC$.

Step 4. Mark where the curves cross. Drop dashed lines to both axes and label the equilibrium price P_m and quantity Q_m.

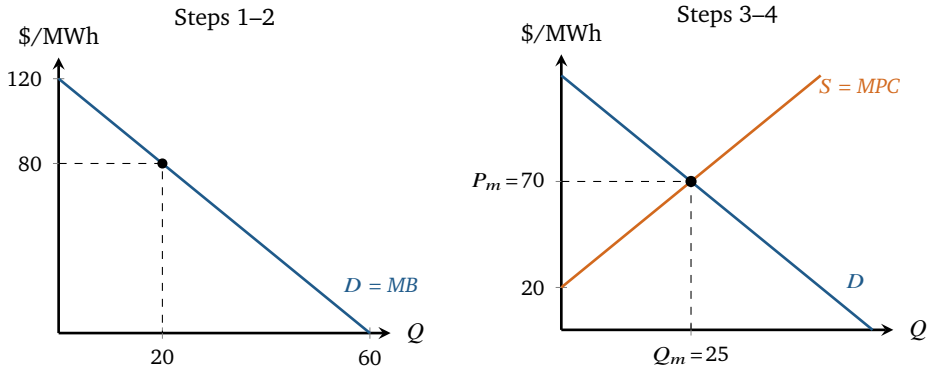


Figure 1.6 — Demand and supply in a regional electricity market, built in steps. Left: demand, read horizontally (at $80 per MWh, buyers want 20 thousand MWh a day) and vertically (the 20th unit is worth $80 to its buyer). Right: supply added; equilibrium at $Q_m = 25$ thousand MWh a day and $P_m = \$70$ per MWh. *(Stylized illustration; not drawn to data.)*

Worked Example 1.6: Solving for market equilibrium

With demand $P = 120 - 2Q$ and supply $P = 20 + 2Q$, the market clears where the last buyer's willingness to pay equals the last seller's marginal cost:

$$120 - 2Q = 20 + 2Q \implies 100 = 4Q \implies Q_m = 25.$$

Substituting into demand, $P_m = 120 - 2(25) = 70$; supply gives the same, $20 + 2(25) = 70$.

Why an equilibrium? At $80, buyers want 20 and sellers offer 30, a

surplus of 10 that pushes the price down. At \$60, buyers want 30 and sellers offer 20, a shortage that pushes the price up. Only at \$70 do the two plans match.

At equilibrium there are no shortages and no surpluses. At $Q_m = 25$ the benefit of the last unit, read off demand, equals its cost, read off supply. That equality, $MB = MC$, is the condition for the efficient quantity, and the next section makes it precise.

Definition: Market equilibrium

The price and quantity at which quantity demanded equals quantity supplied, so no buyer or seller wants to change plans.

1.4.3 Shifts and movements along a curve

Each curve holds many things constant. A change in the good's own price moves the market along a curve; a change in anything else shifts the whole curve. Table 1.1 lists the usual shifters with an environmental example of each. A pollution tax shifts supply up; a heat wave shifts demand out.

Key Point

A change in the good's own price moves you along a curve. A change in anything else shifts the curve.

Table 1.1 — What shifts demand and supply, with an environmental example of each.

Demand shifter	Example	Supply shifter	Example
Income	Rising incomes raise demand for air conditioning and power	Input prices	Cheaper natural gas lowers generators' costs
Prices of related goods	Cheaper heat pumps raise demand for electricity	Technology	Cheaper solar panels shift supply out
Tastes and information	Concern about climate raises demand for green power	Taxes and regulation	A tax per ton of CO_2 shifts supply up
Expectations	Expected price spikes raise demand for home batteries	Expectations	Expected carbon rules deter new coal plants
Number of buyers	New data centers add buyers	Number of sellers	New wind farms enter

Note: Each shifter moves the whole curve; a change in the good's own price moves the market along the curve. Examples are illustrative.

Common Pitfall

"The price rose, so demand fell" is a classic exam error. A higher price lowers the *quantity demanded*, a movement along the demand curve. "Demand fell" means buyers want less at every price, as when a substitute

becomes cheaper.

1.5 The Micro 101 Toolkit II: Surplus, Efficiency and Deadweight Loss

E340 Lecture 1 (micro recap)

Equilibrium describes what a market does. Judging whether the outcome is good requires a measure of how much everyone gains, and that measure is *surplus*, which appears on the graph as an area.

Micro 101 Refresher: Surplus is an area

Consumer surplus (CS) is what buyers would have paid minus what they pay: the area below demand and above the price, out to the quantity sold. *Producer surplus* (PS) is what sellers receive minus their marginal cost: the area above supply and below the price. *Total surplus* is CS + PS, the area between demand and supply up to the quantity traded. With straight-line curves the areas are triangles, and a triangle's area is $\frac{1}{2} \times$ base $\times$ height.

Consumer surplus is a real gain. The buyer at the top of the demand curve would have paid \$120 for her MWh and pays the market price of \$70, so she keeps \$50. Each later buyer keeps a little less, down to the last buyer, who pays exactly what the unit is worth to him.

Producer surplus works the same way from the seller's side. The cheapest generator produces a MWh for about \$20 and sells it for \$70, keeping \$50; the last generator just covers its cost.

Definition: Consumer surplus (CS)

What buyers are willing to pay minus what they pay, summed over all units bought: the area under demand and above the price.

Definition: Producer surplus (PS)

The price sellers receive minus their marginal cost, summed over all units sold: the area above supply and below the price.

Build the Graph Step by Step: Consumer surplus, producer surplus and deadweight loss

Step 1. Start from the finished equilibrium graph of Figure 1.6: demand, supply and the equilibrium at $(Q_m, P_m) = (25, 70)$.

Step 2. Shade consumer surplus: the triangle with corners at the demand intercept (\$120), the price (\$70) and the equilibrium point. Label it CS.

Step 3. Shade producer surplus: the triangle between the price (\$70), the supply intercept (\$20) and the equilibrium point. Label it PS. The two triangles together are total surplus.

Step 4. Push the quantity to $Q = 30$. For each unit from 25 to 30, read the cost off supply and the benefit off demand: cost exceeds benefit. Shade the triangle between the curves from 25 to 30. That lost surplus is the *deadweight loss*.

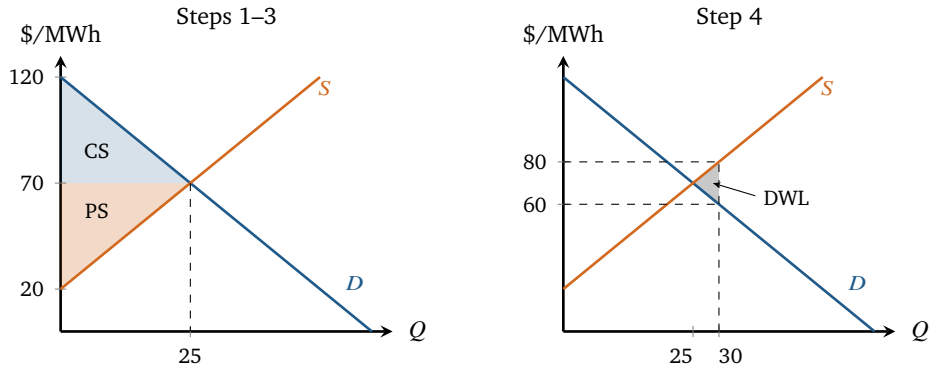


Figure 1.7 — Surplus as area in the electricity market of Figure 1.6. Left: consumer surplus (blue) and producer surplus (orange) at the competitive equilibrium; together they are total surplus. Right: at an output of 30 thousand MWh, each unit beyond 25 costs more to produce (read off S) than it is worth to buyers (read off D); the gray triangle is the deadweight loss. *(Stylized illustration; not drawn to data.)*

Worked Example 1.7: Computing surplus and deadweight loss

Demand $P = 120 - 2Q$, supply $P = 20 + 2Q$, equilibrium $(25, 70)$. Quantities are thousand MWh a day, so areas are thousand dollars a day.

1. CS $= \frac{1}{2} \times 25 \times (120 - 70) = 625$; PS $= \frac{1}{2} \times 25 \times (70 - 20) = 625$; total surplus = 1,250.
2. *Too much.* At $Q = 30$ the last unit is worth $120 - 60 = \$60$ and costs $20 + 60 = \$80$. DWL $= \frac{1}{2} \times (30 - 25) \times (80 - 60) = 50$, or 4 percent of total surplus.
3. *Too little.* At $Q = 15$ the last unit is worth \$90 and costs \$50. DWL $= \frac{1}{2} \times (25 - 15) \times (90 - 50) = 200$, or 16 percent.

Producing more or less than 25 shrinks total surplus. In that sense the competitive quantity is *efficient*.

1.5.1 What "efficient" means

Economists use *efficient* in two related senses. In the first, a quantity is efficient if it makes total surplus as large as possible; any other quantity creates a deadweight loss, value that no one receives.

Definition: Deadweight loss (DWL)
The fall in total surplus when the quantity produced differs from the efficient quantity: value that no one receives.

The second sense is named after the economist Vilfredo Pareto. An allocation is *Pareto efficient* if no change can make someone better off without making someone else worse off; a change that helps someone and hurts no one is a *Pareto improvement*. Phaneuf and Requate (2016) note that the Pareto test appeals because it never says whose well-being counts more. The same feature is its weakness: many outcomes most people would call unfair pass the test.

Definition: Pareto efficiency
An allocation in which no one can be made better off without making at least one other person worse off.

Almost every real policy hurts someone. A pollution rule helps people who breathe cleaner air and costs the coal plant's owners. Economists therefore ask a weaker question: do the winners gain more than the losers lose? If they do, the winners could compensate the losers, and the change is a potential Pareto improvement. Benefit–cost analysis, in Part III, rests on this test.

Definition: Potential Pareto improvement
A change whose winners gain enough that they could fully compensate the losers and still be better off, even if they do not. Also called the Kaldor–Hicks test.

1.5.2 When markets are efficient

The *First Fundamental Theorem of Welfare Economics* states that, under a set of standard assumptions, a competitive equilibrium is Pareto efficient. Buyers and sellers, each pursuing their own interest and guided only by prices, reach an allocation with no free lunches left; in the electricity example the market quantity is the surplus-maximizing quantity. The four assumptions below are the standard introductory summary; they are sufficient conditions, and the formal theorem adds further technical assumptions:

Definition: First Welfare Theorem
Under perfect competition, perfect information, complete markets and no externalities (plus technical assumptions), every competitive equilibrium is Pareto efficient.

1. **Perfect competition:** many buyers and sellers, none able to set the price.
2. **Perfect information:** buyers and sellers know the price and quality of what they trade.
3. **Complete markets:** every good people care about is traded, and making deals is cheap.
4. **No externalities:** every cost and benefit of a trade falls on the buyer or the seller.

Environmental economics works where conditions 3 and 4 fail. No market sold the residents of Cleveland a cleaner river, and no price made the polluters

Key Point
The First Welfare Theorem is the benchmark. Environmental economics studies what happens when its conditions fail.

pay for the trestles their waste set on fire. Phaneuf and Requate (2016) use the welfare theorems the same way: as a checklist for finding where markets go wrong, not as an argument for leaving markets alone.

1.5.3 Efficiency is not fairness

Efficiency concerns the size of the pie; *equity* concerns how it is sliced. If one person owned everything in the world, the allocation would be Pareto efficient, since helping anyone else would mean taking from the owner, yet almost no one would call it fair.

Definition: Equity

Fairness in how benefits and costs are shared across people, places or generations. Efficiency and equity are separate criteria.

Environmental policy raises equity questions constantly. Polluting plants have often been placed in low-income communities and communities of color (Banzhaf et al., 2019). A gasoline tax takes a larger share of income from rural families who drive long distances. The *Second Welfare Theorem* states that, in principle, society can reach any efficient outcome it prefers by redistributing income and then letting markets work. In practice, redistribution is hard, which is why the question "who gains and who pays?" recurs throughout the book and fills Part VI.

Example

A carbon tax raises energy prices for everyone. Returning the revenue as equal per-person rebates can leave most low-income households better off: it changes who pays, not the efficiency gain.

1.6 Why Markets Fail for the Environment: A First Look

E340 Lecture 2 (preview)

The benchmark market of the last two sections gets the quantity right. Environmental markets go wrong for four reasons, previewed here and developed in Chapter 2: externalities, public goods, common-pool resources and missing information.

1.6.1 Externalities

An *externality* is a cost or benefit that a trade imposes on a third party who did not agree to it and is not paid for it. When a coal plant sells a MWh, the deal is between the buyer and the plant. The plant's smoke also reaches people downwind, who pay in coughs, hospital visits and dirty windows.

Definition: Externality

An effect of one agent's action on another's well-being or production that no price reflects and no one compensates. It can be negative (pollution) or positive (pollination).

Phaneuf and Requate (2016) follow the classic definition of Baumol and Oates. An externality exists when one person's well-being or production depends directly on *real* choices made by someone else, with no compensation; "real" means physical things such as smoke or nitrate in groundwater. A new

Definition: Pecuniary effect

An effect that works only through prices, such as a new factory bidding up local wages. It moves money between people but is not an

factory that bids up local wages affects other employers through a price, not through smoke, and is not an externality.

Externalities split private cost from social cost. The power plant's supply curve reflects its *private* costs: fuel, labor and equipment, the *marginal private cost MPC*. The cost to society of the same MWh also includes the damage to third parties, the *marginal external cost MEC* (some instructors write EMC). The two together are the *marginal social cost*:

$$MSC = MPC + MEC. \tag{1.2}$$

Definition: Marginal social cost (MSC)

The full cost to society of one more unit: private cost plus the external cost imposed on others, $MSC = MPC + MEC$.

Micro 101 Refresher: Efficiency means social marginal benefit = social marginal cost

The efficient quantity of any good is where the marginal benefit of the last unit to society equals its marginal cost to society. A competitive market sets private marginal benefit equal to private marginal cost ($MPB = MPC$). With no externalities the private and social curves coincide and the market is efficient. With an external cost, $MSC > MPC$ and the market quantity is too large.

Build the Graph Step by Step: Marginal private cost versus marginal social cost

Step 1. Draw the axes (Q and \$ per unit) and the demand curve $D = MPB$. With no external benefits, demand is also the marginal social benefit MSB.

Step 2. Draw the industry supply curve and label it MPC: the costs the firms pay.

Step 3. Mark the market outcome where D crosses MPC. Label it Q_u ("u" for *unregulated*) and its price P_u.

Step 4. Draw MSC above MPC. The vertical gap between them is MEC, the damage per unit to third parties. Here the gap is constant, so the lines are parallel.

Step 5. Mark where D crosses MSC: the efficient quantity Q^*. Check that $Q^* < Q_u$.

Step 6. Shade the triangle between MSC and D from Q^* to Q_u. Every unit in that range costs society more than it is worth. The triangle is the deadweight loss of the externality.

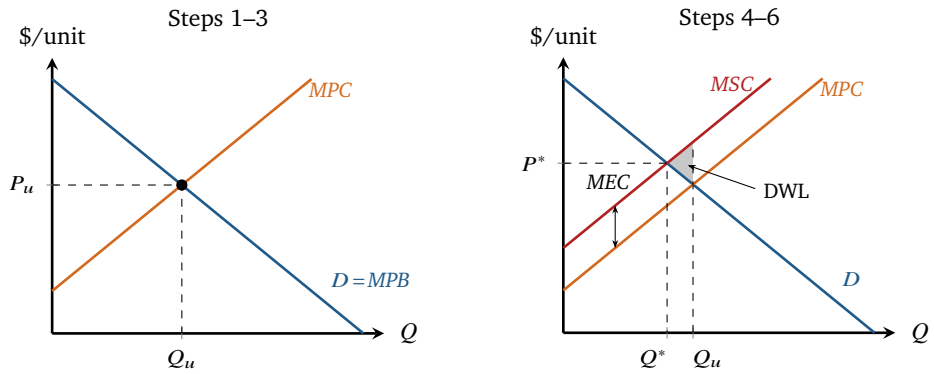


Figure 1.8 — A negative externality, built in steps. Left: the unregulated market, where demand meets marginal private cost at Q_u. Right: adding the marginal external cost gives $MSC = MPC + MEC$; the efficient quantity Q^* is where demand meets MSC. The gray triangle is the deadweight loss from overproduction. *(Stylized illustration; not drawn to data.)*

Worked Example 1.8: The cost of ignoring pollution

Return to the electricity market and suppose each MWh causes \$20 of health and environmental damage downwind (an illustrative number). Then $MEC = 20$ and $MSC = (20 + 2Q) + 20 = 40 + 2Q$.

1. *Market outcome.* Firms ignore the damage: $120 - 2Q = 20 + 2Q$, so $Q_u = 25$ and $P_u = 70$.
2. *Efficient outcome.* Demand equals MSC: $120 - 2Q = 40 + 2Q$, so $Q^* = 20$ and $P^* = 80$.
3. *Deadweight loss.* At $Q_u = 25$, $MSC = 90$ and demand is 70. DWL $= \frac{1}{2} \times (25 - 20) \times (90 - 70) = 50$ thousand dollars a day.
4. *Damage left over.* Total external damage is $20 \times 25 = 500$ at Q_u and $20 \times 20 = 400$ at Q^*. The efficient outcome cuts damage by a fifth; it does not eliminate it.
5. *A fix.* A tax of \$20 per MWh raises MPC by exactly MEC, and the market lands on $Q^* = 20$. Such a *Pigouvian tax* is the subject of Chapter 5.

Common Pitfall

With an external cost, students often shift the demand curve. The external cost is a cost of production, so it belongs on the cost side: draw MSC above MPC. External benefits from consumption, such as

vaccination, go on the demand side: *MSB* above *MPB*.

1.6.2 Public goods and common-pool resources

Many environmental goods have two unusual features. A good is *non-rival* if one person's use does not reduce what is left for others.

Definition: Non-rival, non-excludable

Non-rival: one person's use leaves as much for others. Non-excludable: no one can be kept from using it.

A good is *non-excludable* if keeping people from using it is impossible or too costly. Clean city air has both features. The air's capacity to absorb pollution is a different matter. Breathing clean air leaves as much for others, but every ton one factory emits uses up capacity that is then unavailable to others, so the air used as a dump is rival and behaves like a common-pool resource. Cleaning the air supplies a public good; polluting it draws down a common pool. Crossing the two features gives four kinds of goods (Table 1.2).

Table 1.2 — Four kinds of goods, by rivalry and excludability.

	Excludable	**Non-excludable**
Rival	Private goods (food, gasoline, electricity)	Common-pool resources (fish stocks, groundwater, timber on open land)
Non-rival	Club goods (toll roads, streaming, a gated park)	Public goods (clean air, climate stability, flood control)

Public goods are under-provided because of *free riding*: each person hopes others will pay, since everyone enjoys the good either way (Samuelson, 1954).

Definition: Free rider

Someone who enjoys a good without paying for it. Free riding is why voluntary contributions under-provide public goods.

Common-pool resources open to all have the opposite problem: they tend to be overused. Each boat that takes another ton of fish leaves fewer for everyone else, but no single boat bears that cost; Hardin (1968) calls the result the tragedy of the commons. Elinor Ostrom shows that many communities avoid the tragedy by writing their own rules for shared resources (Ostrom, 1990); Chapter 3 studies her work.

Example

Groundwater is a common pool. Each farmer pumping from a shared aquifer lowers the water table under every neighbor, but pays only for his own pumping.

1.6.3 Missing information

Markets also fail when people do not know the risks they face. Lead in tap water is invisible, as Flint's residents learned. People who cannot see a risk cannot protect themselves or demand protection. Governments respond

Definition: Imperfect information

Buyers or sellers lack knowledge of prices, quality or risks, so they cannot choose in their own best interest.

with air-quality alerts, drinking-water reports and labels; Chapter 9 and Chapter 14 ask whether such information changes behavior, and for whom.

1.6.4 Two ways to fix a market failure

Phaneuf and Requate (2016) describe two traditions that bracket most debates over environmental policy. The *Pigouvian* tradition, after Arthur Cecil Pigou, treats an externality as a reason for government to act, for example with a tax equal to the marginal damage (Pigou, 1920).

Definition: Pigouvian tax

A tax per unit of a harmful activity set equal to the marginal external damage at the efficient level, so private decision makers face the full social cost.

The *Coasean* tradition, after Ronald Coase, treats it as a problem of unclear property rights: if rights are clear and bargaining is cheap, the affected parties can negotiate their way to the efficient outcome (Coase, 1960). Real policies sit between the two, and Parts I and II move back and forth between them.

Trace It: India and Brazil — Why problems arise: missing property rights

Institution

U.S.	The Clean Water Act of 1972 created a permit program for point sources (U.S. Environmental Protection Agency, 2026a); before it, a river could serve as a cheap sewer.
India	Indian law lets every landowner collect and dispose of the water under the land (Government of India, 1882).
Brazil	Rivers and aquifers are public property (Brasil, 1988, 1997); much frontier land is untitled, and one federal program titles land in "productive use" (Baragwanath and Shinde, 2026).

Rule or market

U.S.	Discharge permits; for SO_2, a market in allowances (Chapter 6).
India	No price on pumping: farm power is often free or flat-rate, and some states ration hours of supply (Ryan and Sudarshan, 2022).
Brazil	Forest Code: Amazon forest properties must, as a rule, keep 80 percent in native vegetation (Brasil, 2012) and register in the self-declared CAR, which confers no title (Shinde et al., 2027).

Data

U.S.	EPA permit and discharge records.
India	Observation wells (Central Ground Water Board and state groundwater agencies, 2026); satellite gravity.
Brazil	Clearing maps; CAR and land-registry boundaries (Baragwanath and Shinde, 2026).

Evidence and difference

U.S.	Water pollution has fallen since 1972, partly because of the law (Keiser and Shapiro, 2019).
India	Rationing holds average pumping near the efficient level but gives water to the wrong farmers (Ryan and Sudarshan, 2022).
Brazil	Titling for productive use raises clearing before title (Baragwanath and Shinde, 2026); in Pará, early CAR registration cut land-conflict events by about 12 percent, most where enforcement capacity was stronger (Shinde et al., 2027).

1.7 The Efficient Level of Pollution Is Not Zero

E340 Lectures 1 and 4

The efficient amount of electricity in the last section still comes with pollution. Putting pollution itself on the horizontal axis makes the point directly, in the graph you will draw more than any other in E340.

Micro 101 Refresher: Marginal thinking and the equimarginal rule

To choose the best level of any activity, compare the marginal benefit of one more unit with its marginal cost; total net benefit is largest where the two are equal. For pollution, the benefit of allowing one more ton is the abatement cost avoided, and the cost of that ton is the damage it does.

Two curves do the work. The *marginal abatement cost* curve, *MAC*, shows what it costs to cut emissions by one more ton. With no regulation, a firm emits E_0. The first tons it removes are cheap (better maintenance, cleaner fuel) and each later ton costs more. Reading right to left from E_0, costs climb, so *MAC* falls from left to right and reaches zero at E_0.

The *marginal damage* curve, *MD*, shows the harm from one more ton. In clean air an extra ton does little harm; in dirty air it pushes more people past health thresholds. *MD* therefore usually rises with emissions.

Definition: Marginal abatement cost (MAC)

The cost of cutting emissions by one more unit, starting from a given level. With emissions on the x-axis, MAC slopes down.

Definition: Marginal damage (MD)

The extra harm (health, property, ecosystems, climate) caused by one more unit of emissions. Usually rising: a ton does more harm when the air is already dirty.

Build the Graph Step by Step: The efficient level of pollution (*MAC* = *MD*)

Step 1. Draw the axes: emissions E (tons) on the horizontal axis, \$ per ton on the vertical axis. Mark E_0, the unregulated emissions level, on the horizontal axis.

Step 2. Draw *MAC* falling from the vertical axis to zero at E_0. Read it right to left from E_0: the first tons abated are cheap, the last ones

expensive.

Step 3. Draw *MD* rising from the origin.

Step 4. Mark where the curves cross and drop a dashed line to the axis: that is E^*, the efficient level of emissions. Shade total damages (under *MD* from 0 to E^*) and total abatement cost (under *MAC* from E^* to E_0).

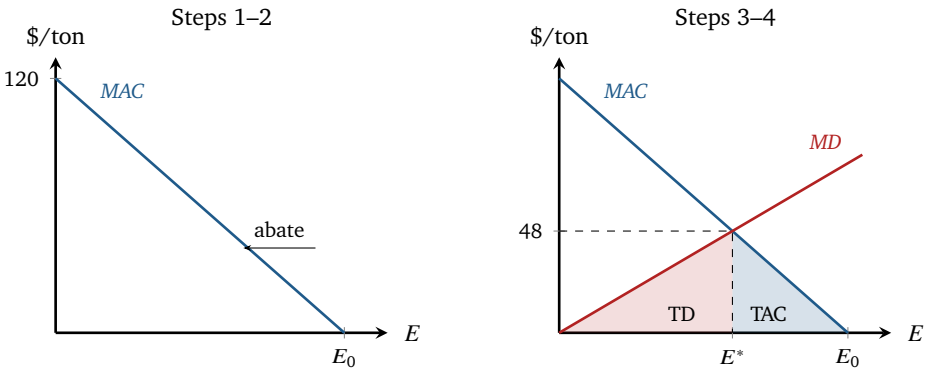


Figure 1.9 — The efficient level of pollution, built in steps, with emissions E on the horizontal axis. Left: *MAC* falls to zero at unregulated emissions E_0; abatement means moving left. Right: rising *MD* added; the crossing gives E^*. Red area: total damages (TD) at E^*. Blue area: total abatement cost (TAC) at E^*. *(Stylized illustration; not drawn to data.)*

Worked Example 1.9: Finding the efficient level of emissions

A U.S. paper mill on the White River emits $E_0 = 40$ tons of a pollutant a week. Its marginal abatement cost is $MAC = 120 - 3E$ and the marginal damage downstream is $MD = 2E$, in dollars per ton (illustrative numbers).

1. *Efficient level.* $120 - 3E = 2E$, so $E^* = 24$ tons, where $MAC = MD =$ \$48 per ton. The mill abates $40 - 24 = 16$ tons, or 40 percent of its emissions.
2. *Total damages at* E^*: $\frac{1}{2} \times 24 \times 48 =$ \$576.
3. *Total abatement cost at* E^*: $\frac{1}{2} \times (40 - 24) \times 48 =$ \$384.
4. *Total social cost*: $576 + 384 =$ \$960.

Compare the extremes. With no regulation ($E = 40$), abatement costs nothing but damages are $\frac{1}{2} \times 40 \times 80 =$ \$1,600. With zero pollution ($E = 0$), damages are zero but abatement costs $\frac{1}{2} \times 40 \times 120 =$ \$2,400, two and a half times the cost at E^*. The best outcome lies between,

with 24 tons of pollution.

The efficient level of pollution is usually not zero. Removing the last tons would cost far more than the harm they do, and the money could do more good elsewhere. The result does not make pollution good; it says that avoiding pollution has an opportunity cost.

Key Point

Efficient pollution sets $MAC = MD$. It is zero only if even the first ton does more damage than it costs to remove.

Common Pitfall

Two common errors. (1) Drawing *MAC* rising: with *emissions* on the x-axis *MAC* falls, and with *abatement* on the x-axis it rises, so check the axis label first. (2) Placing the efficient level where total damages equal total abatement cost: E^* is where the *marginal* curves cross. In the worked example the areas at E^* are \$576 and \$384.

1.8 Evidence: Which Explanations Survive

E340 Lecture 1

Four popular claims about pollution and growth can be tested against the facts of Section 1.1. The first two fail; the last two are rejected in the other direction, by evidence that clean air is valued and that U.S. air rules pass a benefit–cost test.

"Growth cleans up the environment by itself." In an influential study, Grossman and Krueger (1995) find that, for several local air and water pollutants, pollution first rises and then falls as national income rises, tracing an inverted U. The pattern became known as the *environmental Kuznets curve* (Figure 1.10). Later work is skeptical: Stern (2004) shows that many such results depend on the statistical method and often fail for CO_2.

Definition: Environmental Kuznets curve (EKC)

A proposed inverted-U relationship between income per person and pollution: pollution first rises with income, then falls once countries are rich enough.

The data of Figure 1.1 fit the skeptical reading. U.S. SO_2 fell 95 percent as income grew, but it fell after laws capped it; CO_2, which no national law priced, did not fall much. Rising income makes cleanup affordable and raises the demand for it, and policy turns that demand into lower emissions. The public-goods logic of Section 1.6 explains the rest: a country that cuts CO_2 bears the whole cost and gets a small share of the benefit, while local SO_2 cuts benefit the voters who pay for them.

Key Point

Where pollution falls with income, it is usually because richer societies choose stricter policy, not because growth cleans the air by itself.

"Fossil fuels will run out and solve the problem." Covert et al. (2016) check the record. For about three decades, proven world reserves of oil

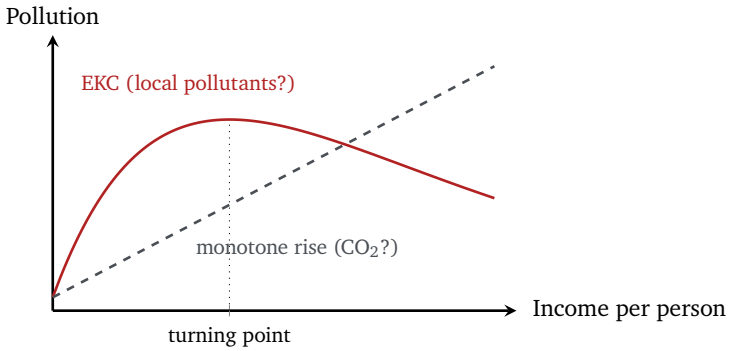


Figure 1.10 — The environmental Kuznets curve hypothesis (solid) and a pollutant that keeps rising with income (dashed). Which pattern a pollutant follows, and why, is an empirical question. *(Stylized illustration; not drawn to data.)*

and natural gas have stayed at roughly 50 years of current use, because new discoveries keep pace with consumption. They conclude that, without policies that price the external costs of fossil fuels, market forces alone are unlikely to cut their use by much.

"People do not value clean air enough to pay for it." House prices reject the claim, as the research box below reports.

From the Research: Do people value clean air? Chay and Greenstone (2005)

Chay and Greenstone (2005) use the 1970 Clean Air Act as a natural experiment. Counties whose particulate pollution exceeded the federal standard were labeled "nonattainment" and faced stricter regulation. Pollution fell faster in those counties, and house prices rose faster. Because the label came from a threshold rule rather than from local economic trends, the comparison works like an experiment. The authors estimate that a 1 percent fall in particulate concentrations raises house values by 0.20 to 0.35 percent, and they associate the regulation with an increase of about $45 billion in housing values in nonattainment counties between 1970 and 1980. Housing prices reveal a willingness to pay for clean air; Chapter 13 develops this *hedonic* method.

"Environmental rules cost more than they are worth." For U.S. air rules the official estimates point the other way. EPA's prospective study of the 1990 Clean Air Act Amendments projects that by 2020 the amendments prevent

over 230,000 early deaths a year, with direct benefits of almost $2 trillion against direct costs of about $65 billion (both in 2006 dollars), a ratio above 30 to 1 (U.S. Environmental Protection Agency, 2011). The projections rest on a model of what would have happened without the law, and most of the benefits are avoided deaths valued in dollars. The next section takes up that valuation.

1.9 Policy, and the Road Ahead

Governments have five main tools for environmental problems: *regulations* that require or forbid actions (an emissions limit, a required technology); *taxes and fees* that make harmful behavior more expensive; *tradable permits* (cap and trade) that fix total pollution and let firms buy and sell the right to emit; *subsidies* that reward cleaner technologies; and *information* that helps people see and avoid risks. Market-based tools, taxes and permits, can often reach a goal at lower cost than rigid rules, because they let the firms that can cut pollution cheaply do more of the cutting (Keohane and Olmstead, 2016).

E340 judges a policy on four criteria: *effectiveness* (does it improve the environmental outcome?), *efficiency* (are the benefits worth the costs, and is the goal reached at low cost?), *distribution* (who gains and who pays?) and *feasibility* (can it be implemented and enforced?). Benefit–cost analysis serves the second criterion. A group of leading economists argues that it should inform environmental, health and safety decisions without being the only test (Arrow et al., 1996).

Feasibility is where the three countries differ most: a rule works only if someone can monitor and enforce it.

Trace It: India and Brazil — Who acts: agencies, courts and prosecutors

Institution	
U.S.	EPA and state agencies (U.S. Environmental Protection Agency, 2026d).
India	MoEFCC, the CPCB and state boards; since 2010, the National Green Tribunal, an environmental court (Government of India, 2010).
Brazil	SISNAMA, the system of federal, state and municipal agencies (Brasil, 1981); IBAMA (1989) holds federal environmental police power (Brasil, 1989).

India and Brazil

EPA. India: Ministry of Environment, Forest and Climate Change (MoEFCC), the CPCB and state pollution control boards. Brazil: Ministry of the Environment (MMA), IBAMA, ICMBio and state agencies.

E340 Lecture 1

Definition: Cost-effective

Reaching a given environmental goal at the lowest total cost. A policy can be cost-effective without being efficient, if the goal itself is set at the wrong level.

Key Point

Benefit–cost analysis informs decisions; it does not replace them. It makes the trade-offs visible.

Rule or market	
U.S.	Statutes carried out through standards and permits (Part V).
India	Statutes plus courts: the Supreme Court made hazardous industries absolutely liable for harm (Supreme Court of India, 1986), and the Tribunal's act ties a healthy environment to the right to life (Government of India, 2010).
Brazil	A constitutional right to an ecologically balanced environment (Brasil, 1988); polluters repair damage regardless of fault, and public prosecutors (Ministério Público) may sue (Brasil, 1981, 1985).
Data	
U.S.	Enforcement records; benefit–cost studies (U.S. Environmental Protection Agency, 2011).
India	Board inspections and audits; court orders.
Brazil	IBAMA fines and embargoes; DETER alerts.
Evidence and difference	
U.S.	Stricter rules in counties above the particle standard cut pollution and raised house values (Chay and Greenstone, 2005).
India	Air rules are associated with substantial gains in air quality, water rules with no measurable benefits; public demand may explain the gap (Greenstone and Hanna, 2014).
Brazil	Satellite-guided enforcement curbs clearing (Assunção et al., 2023), and policy, not only prices, explains much of the decline after 2004 (Assunção et al., 2015).

1.9.1 Counting lives saved

The value of avoided deaths became a policy dispute in 2026. In January of that year, *The New York Times* reported that EPA would stop putting a dollar value on the health benefits of cutting fine particles ($PM_{2.5}$) and ozone in new air rules, while continuing to count the rules' costs to industry (Joselow, 2026). EPA's leadership objected to the report and said the agency would still *consider* health effects; what it would stop doing was *monetizing* them.

The tools of this chapter split the question into three parts. The positive part asks how many deaths a rule prevents and what it costs. The measurement part asks how much people are willing to pay for a small reduction in their risk of dying, the subject of Chapter 12. The normative part asks how uncertain benefits should be weighed against more certain costs, and who should pay. On the graph of Figure 1.9, dropping the value of avoided deaths erases most of *MD* and moves the computed "efficient" level of pollution far

Key Point

Leaving a benefit out of an analysis is not neutral: it values that benefit at zero.

to the right, toward E_0. Economics does not settle the dispute, but it shows what is at stake.

1.9.2 How this book is organized

The book follows the order of E340 in six parts (Table 1.3), with the three threads of finance, water and climate running through all of them.

Table 1.3 — The six parts of the book, their chapters, and the E340 lectures each part draws on.

Part	Chapters	E340 lectures
I. Foundations	1 Introduction; 2 Market failures; 3 Property rights, Coase and Ostrom	1–3, 5
II. Policy Instruments	4 Standards; 5 Taxes and subsidies; 6 Tradable permits; 7 Second best and uncertainty	4 (+ supp.)
III. Measuring Benefits and Costs	8 Econometrics; 9 Environmental data; 10 Discounting and benefit–cost analysis; 11 Travel cost; 12 Stated preference; 13 Hedonics; 14 Defensive behavior	6–12
IV. Climate Change, Carbon and Finance	15 Climate economics; 16 Carbon offsets; 17 Climate finance; 18 Natural capital	13–15
V. U.S. Environmental Regulation	19 Economics of regulation; 20 Clean Air Act; 21 Clean Water Act; 22 Safe Drinking Water Act; 23 Hazardous waste; 24 Transportation	18–23
VI. Resources, Justice and the Road Ahead	25 Forests and fisheries; 26–27 Environmental justice (U.S. and global); 28 Market-based policy in practice; 29 What's next	24–28

Note: Lecture numbers refer to the Fall 2026 E340 schedule; "supp." marks supplementary lectures.

Each part teaches through U.S. cases and traces them to India and Brazil: India's free farm power (Chapter 2) and groundwater (Chapter 18, Chapter 27); Amazon land rights, enforcement and finance (Chapter 3, Chapter 4, Chapter 17, Chapter 25); subsidies and fuel taxes in both countries (Chapter 5); and India's wells beside Brazil's satellites (Chapter 9).

1.10 Same Idea, Three Settings

The economics of this chapter is the same in all three countries: a resource that is hard to fence off, and whose use leaves less for others, is overused when no price, owner or shared rule limits its use. Industries on the Cuyahoga used a river as a sewer, farmers in northwest India pump an aquifer that an 1882 law lets any landowner draw from, and landholders on the Amazon frontier clear forest that stores carbon for the world. Each is an externality or common-pool problem (Section 1.6) with an efficient level set by $MAC = MD$.

What differs is who owns the resource, who can measure its use and who can enforce a rule. The United States assigned discharge rights through permits and built long records. India has pollution statutes and an active Supreme Court, but monitoring is costly and can be corrupted, and free farm power blocks a price on groundwater (Duflo et al., 2013; Ryan and Sudarshan, 2022). Brazil sees its forests from space and has used that to enforce the law, but frontier land rights are unsettled, so the rules for gaining title can reward clearing (Assunção et al., 2023; Baragwanath and Shinde, 2026).

Income and public demand matter too: Greenstone and Hanna (2014) suggest that demand for clean air explains why India's air rules worked better than its water rules. In each chapter, ask the trace-box questions: who decides, under which rule, with what data, and why the result differs.

Key Point

The model travels unchanged across countries. The institutions that carry it out do not, and they explain most of the differences in results.

Definition: Enforcement capacity

The ability of a state to detect violations and impose penalties: inspectors, monitors, courts and the budget to use them. A rule with weak enforcement changes behavior little.

Chapter Summary

- Since 1970, U.S. real GDP has grown 4.5-fold while SO_2 emissions fell 95 percent; U.S. CO_2 is 13 percent above 1970, and world CO_2 has risen 2.6-fold to 38.6 Gt.
- Environmental problems are allocation problems: scarce environmental resources used without a price are overused.
- Scarcity, opportunity cost, marginal thinking and incentives are the core tools. On the PPF, more environmental quality costs other goods; inside it, better policy gives more of both.
- Nature is an asset. Its value is the present value of its services, and degrading it is depreciation. Weak sustainability keeps total capital intact; strong sustainability protects critical natural capital.

- Demand is marginal benefit and supply is marginal cost. Surplus is an area, and a competitive market maximizes total surplus when the First Welfare Theorem's conditions hold.
- Efficiency (the size of the pie) and equity (how it is sliced) are separate criteria.
- An external cost makes $MSC = MPC + MEC$ exceed private cost, so the market overproduces and creates a deadweight loss. Public goods are under-provided, and common-pool resources open to all are overused.
- The efficient level of pollution sets $MAC = MD$ and is usually not zero.
- The evidence does not support "growth cleans up by itself" or "fossil fuels will run out"; it shows that people pay for clean air and that U.S. air rules have projected benefits far above their costs.
- Policies are judged on effectiveness, efficiency, distribution and feasibility.
- India's largest environmental problems are air pollution (1.67 million deaths attributed to it in 2019) and groundwater depletion; Brazil's is forest clearing, which makes up about 77 percent of its CO_2 emissions.
- Traced to India and Brazil, the same economics applies, but results differ with who owns the resource, the cost of monitoring and the capacity to enforce: India relies on courts alongside regulators whose data can be biased; Brazil enforces with satellites but has unsettled land rights on its frontier.

Key Terms

- Scarcity
- Opportunity cost
- Marginal benefit, marginal cost
- Production possibilities frontier
- Positive and normative analysis
- Natural capital
- Ecosystem services
- Present value
- Willingness to pay
- Consumer surplus
- Producer surplus
- Deadweight loss
- Pareto efficiency
- First Welfare Theorem
- Equity
- Externality

- Marginal social cost
- Public good; common-pool resource
- Marginal abatement cost
- Marginal damage
- Environmental Kuznets curve
- Weak and strong sustainability
- Stock and flow; depletion
- Land-use change emissions
- Enforcement capacity
- Trace (U.S., India, Brazil equivalents)

Exercises

1.1 [Basic] Explain the opportunity cost of (a) keeping a 500-acre forest near a growing suburb, and (b) cutting it down for housing. Who bears each cost?

1.2 [Basic] Positive or normative? Explain. (a) "The 1990 Clean Air Act Amendments reduced SO_2 emissions." (b) "EPA should value a statistical life at $10 million." (c) "A gasoline tax will reduce driving." (d) "Low-income households should not pay more for clean air than high-income households."

1.3 [Basic] *Draw the graph.* Draw a PPF with "market goods" on the horizontal axis and "water quality in Lake Monroe" on the vertical axis. Mark an efficient point, a wasteful point and an unattainable point. Then show the effect of a cheaper way to control farm runoff.

1.4 [Basic] A county park near Bloomington provides recreation worth $80,000 a year forever. (a) Find its present value at $r = 4\%$ and at $r = 2\%$. (b) A mining company offers $2.5 million for the land. At which rate does selling make sense, counting only recreation? (c) Name two other services the park provides.

1.5 [Intermediate] *Draw the graph.* Demand is $P = 100 - Q$ and supply is $P = 10 + 0.5Q$. (a) Draw both curves with intercepts. (b) Find and mark the equilibrium. (c) Shade and compute CS, PS and total surplus. (d) Compute and shade the deadweight loss if output is capped at $Q = 40$.

1.6 [Intermediate] *Draw the graph.* A U.S. factory has $MPC = 10 + Q$, its pollution causes $MD = 2Q$ per unit of output, and demand is $P = 70 - Q$. (a) Derive MSC. (b) Find Q_u and Q^*. (c) Draw the graph and shade

the deadweight loss. (d) Compute it. (e) Consumer surplus is smaller at Q^* than at Q_u. Is society worse off at Q^*? Explain.

1.7 [Intermediate] *Draw the graph.* A wastewater plant in Ohio discharges phosphorus E (tons a year) into the Maumee River, with $MAC = 90 - 2E$ and $MD = E$. (a) Find E_0 and E^*. (b) Draw the graph; shade total damages and total abatement cost at E^*. (c) Compute total social cost at E^*, E_0 and $E = 0$. (d) Explain in plain words why $E^* > 0$.

1.8 [Intermediate] In Exercise 1.7, new research finds that the phosphorus also feeds toxic algae near a drinking-water intake, as in Toledo in 2014, so $MD = 2E$. (a) Find the new E^*. (b) Show the change on your graph. (c) Relate the result to the 2026 debate over monetizing health benefits.

1.9 [Intermediate] *Draw the graph.* Draw four externality diagrams: (a) negative production (a coal plant); (b) positive production (a beekeeper whose bees pollinate orchards); (c) negative consumption (gas leaf blowers); (d) positive consumption (street trees). Label private and social curves, Q_u and Q^*, shade the deadweight loss, and say whether the market over- or under-provides.

1.10 [Intermediate] Using Figure 1.1 and Figure 1.2: (a) describe what happened to U.S. SO_2 and CO_2 emissions since 1970; (b) give one economic reason SO_2 fell much more; (c) do the figures show that growth by itself solves pollution?

1.11 [Challenge] A country earns \$20 billion a year selling natural gas. GDP is \$200 billion, gross investment \$30 billion, depreciation \$25 billion, and river pollution destroys \$5 billion a year of natural capital. (a) Compute green NNP. (b) Is the country weakly sustainable? (c) How much extra yearly investment does the Hartwick rule require? (d) Would a strong-sustainability advocate be satisfied? Why?

1.12 [Challenge] Write a one-page memo to a regulator on whether the lives saved by an air rule should be counted as a benefit. (a) Separate the positive, measurement and normative parts. (b) Use the *MAC–MD* graph to show what happens to the "efficient" level of $PM_{2.5}$ if the mortality benefits are set to zero. (c) Apply the four policy criteria of

Section 1.9.

1.13 [Basic] *India twin of Exercise 1.4.* A gram panchayat (village council) in Andhra Pradesh owns an irrigation tank that also recharges village wells. Its services are worth Rs 12 lakh (Rs 1,200,000) a year forever, and a builder offers Rs 2 crore (Rs 20,000,000) for the site (illustrative numbers). (a) Find the tank's present value at $r = 4\%$ and at $r = 8\%$. (b) Above what discount rate does the sale make sense, counting only these services? (c) Farmers whose wells the tank recharges are not party to the sale; explain the externality. (d) Compare your break-even rate with that of the U.S. park in Exercise 1.4.

1.14 [Basic] *Brazil twin of Exercise 1.4.* A landholder in Pará owns a hectare of standing forest that yields Brazil nuts and açaí worth R$400 a year forever. Clearing it for pasture yields a timber sale and pasture rents with a present value of R$9,000 (illustrative numbers). (a) Find the present value of the standing forest at $r = 4\%$ and $r = 6\%$. Which use does the landholder choose at each rate? (b) Using the Amazon Fund's convention of 367 tonnes of CO_2 per hectare and $5 a tonne (Worked Example 1.4), what is the carbon value of the hectare in dollars? (c) Who receives that carbon value, and why does it not enter the landholder's choice? (d) Compared with the U.S. park in Exercise 1.4, which case sends more benefits to people outside the sale?

1.15 [Intermediate] *Draw the graph. India twin of Exercise 1.7.* A sugar mill in Maharashtra discharges organic waste, measured as biochemical oxygen demand (BOD), into a river. Unregulated, it discharges $E_0 = 20$ tonnes of BOD a month. Its marginal abatement cost is $MAC = 60 - 3E$ and the marginal damage downstream is $MD = 2E$, both in thousand rupees per tonne (illustrative numbers). (a) Find E^* and the marginal cost and damage there. (b) Draw the graph; shade total damages and total abatement cost at E^*. (c) Compute total social cost at E^*, at E_0 and at $E = 0$. (d) The state pollution control board's consent to operate limits the mill to 10 tonnes a month. Compute total social cost at $E = 10$ and compare it with E^*. (e) Compared with the U.S. plant of Exercise 1.7, give one reason from the trace boxes why the limit might not be met in practice.

1.16 [Intermediate] *Draw the graph. Brazil twin of Exercise 1.6.* In a frontier municipality of the Amazon, the marginal private benefit of clearing land for pasture is $MPB = 3{,}000 - 20Q$ and the marginal private cost of clearing and fencing is $MPC = 500 + 5Q$, where Q is thousand hectares cleared a year and both are in reais per hectare. The carbon released does R$1,500 of damage per hectare to people elsewhere (illustrative numbers). (a) Find the unregulated quantity Q_u and the efficient quantity Q^*. (b) Draw the graph with *MPB*, *MPC* and *MSC*, and shade the deadweight loss. (c) Compute the deadweight loss in reais. (d) IBAMA can fine illegal clearing, but it detects only a share p of it. If $p = 0.25$, what fine per hectare would make the expected fine equal to the external damage? (e) Explain why satellite alerts raise p, and compare with how a U.S. regulator learns of a violation.

1.17 [Intermediate] Using Figure 1.4: (a) Rank the United States, India and Brazil by fossil CO_2 per person in 2024, and again counting land-use change. (b) Explain why Brazil's rank changes and India's and the United States' do not. (c) A carbon tax on fossil fuels is proposed in all three countries. In which country would it reach the smallest share of CO_2 emissions, and what other policy would that country need? (d) Why are the land-use change estimates less certain than the fossil ones?

Further Reading

- Keohane and Olmstead (2016), Chapters 1, 2, 4 and 11. The E340 textbook: efficiency and the equimarginal rule (Ch. 2), market efficiency (Ch. 4) and sustainability (Ch. 11).
- Phaneuf and Requate (2016), Chapters 1–2. The welfare economics behind this chapter (Pareto efficiency, the welfare theorems, externalities), a history of environmental policy, and a survey of air, water and land pollution. More advanced, and worth dipping into.
- Tietenberg and Lewis (2023), Chapters 1–2, and Jaeger (2005): a broad undergraduate survey, and a short case for economic thinking about the environment.
- Solow (1993) and Covert et al. (2016). Two short E340 response-paper readings: sustainability as keeping capital intact, and why fossil fuels will not fade away on their own.
- Stradling and Stradling (2008). How a twenty-minute fire on the Cuyahoga

became a national symbol.

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